

Project Proposal #2

Project Title: Predicting which Hogwarts house Harry Potter Characters belong to.

The data set I am using is called '**Harry Potter Sorting Dataset**' published on Kaggle by SAHITYA PALACHARLA

The link to the dataset: <https://www.kaggle.com/datasets/sahityapalacharla/harry-potter-sorting-dataset/data>

This is a tabular dataset which contains 1000 rows and 10 attributes. Each row represents a character, and the attributes are the following: blood status, bravery, intelligence, loyalty, ambition, dark arts knowledge, quidditch skills, dueling skills, creativity, and house (which is the target attribute)

This dataset fits for my project because based on these 9 variables, I can predict which house each character belongs to. This dataset is interesting because each Hogwarts house has a dominant criterion for accepting students (if a student has a high bravery score they would go to Gryffindor, if they had a high intelligence score they would go to Ravenclaw, high loyalty to Hufflepuff, and high ambition to Slytherin). However, in the books it is made very apparent that Slytherin house is prejudiced to half-bloods and muggle-born, and favor individuals with high dueling ability. Therefore it would be interesting to see how my model would perform to a character who is muggle born but has a high ambition score and is good at dueling. Furthermore, creativity and quidditch skills should have no effect on which house a character gets sorted into, therefore I intend to train a model with all attributes and a model where I drop these two columns and compare the results.

I plan to use numerous classifier algorithms for this project (the algorithms covered in class and the ones I had used for the House Price Prediction homework).